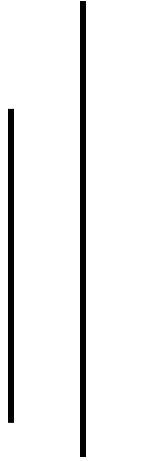


ASSIGNMENT OF OOP (OBJECT-ORIENTED PROGRAMMING)



Tutorial-1

Name: Aashutosh Subedi

Section: "D"

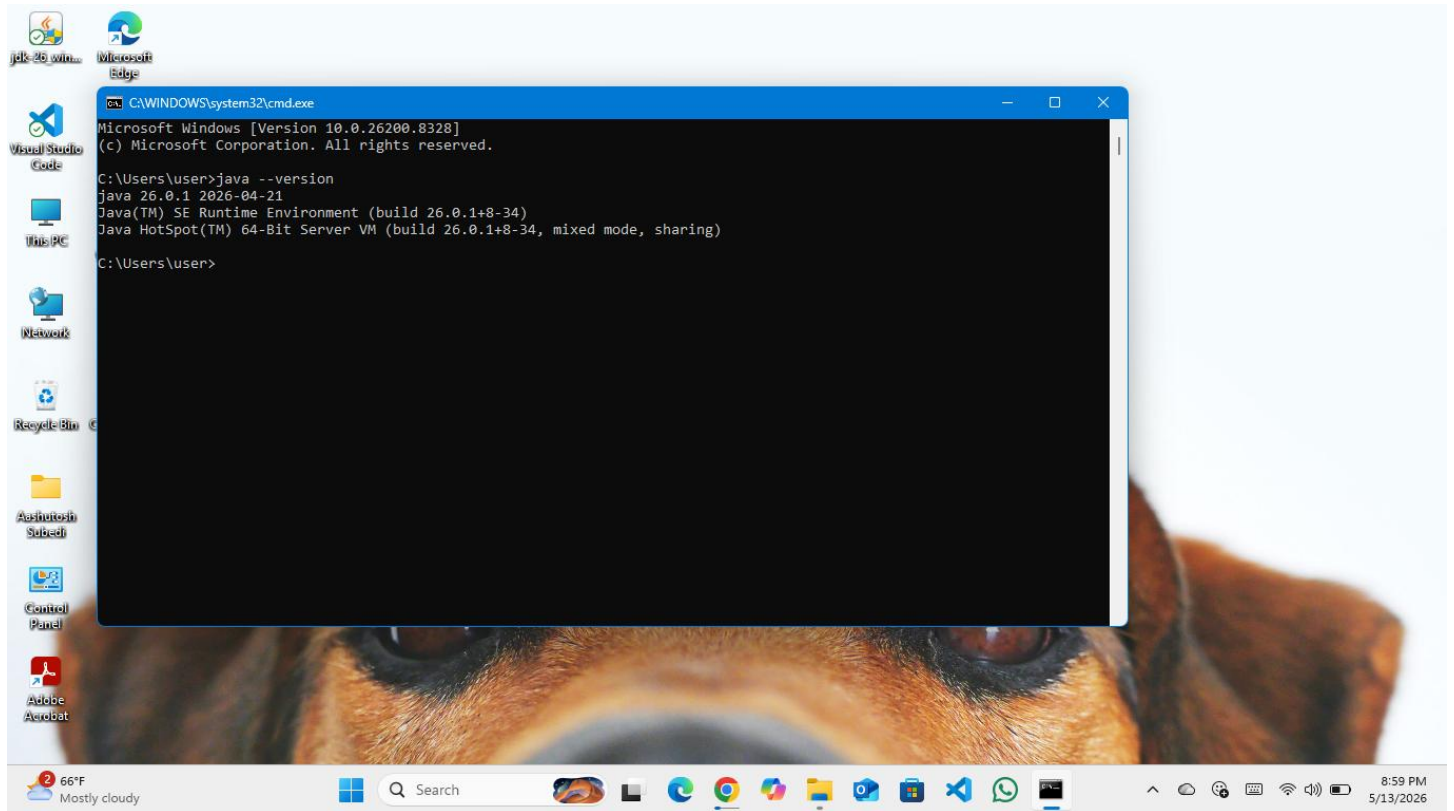
Course: BCSNT 6063- object oriented programming (2 semester)

Submitted to;

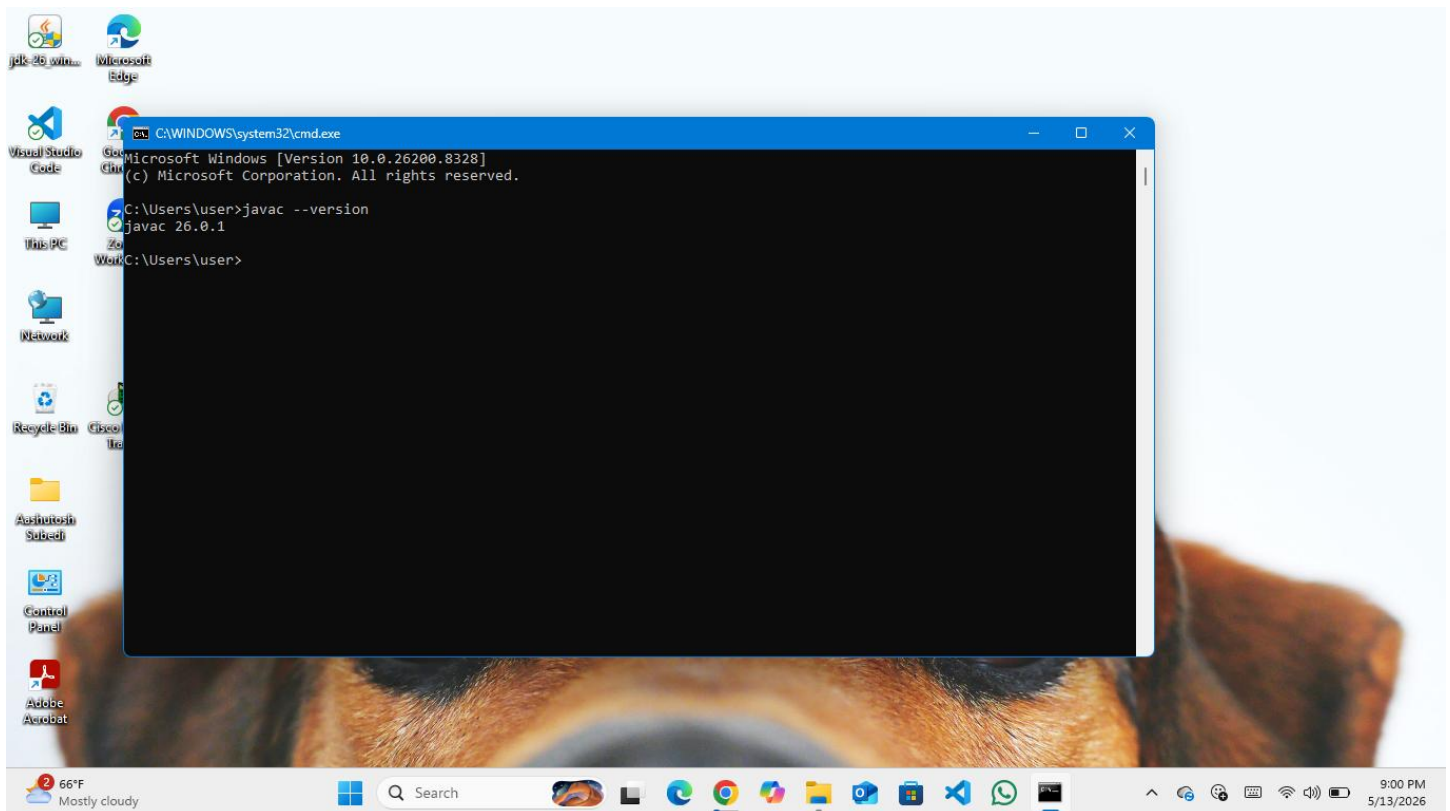
Lecturer: Sanju Shrestha

1. Java Development Kit (JDK) Installation and Verification

i. Screenshot 1: Java version check confirming JDK installation



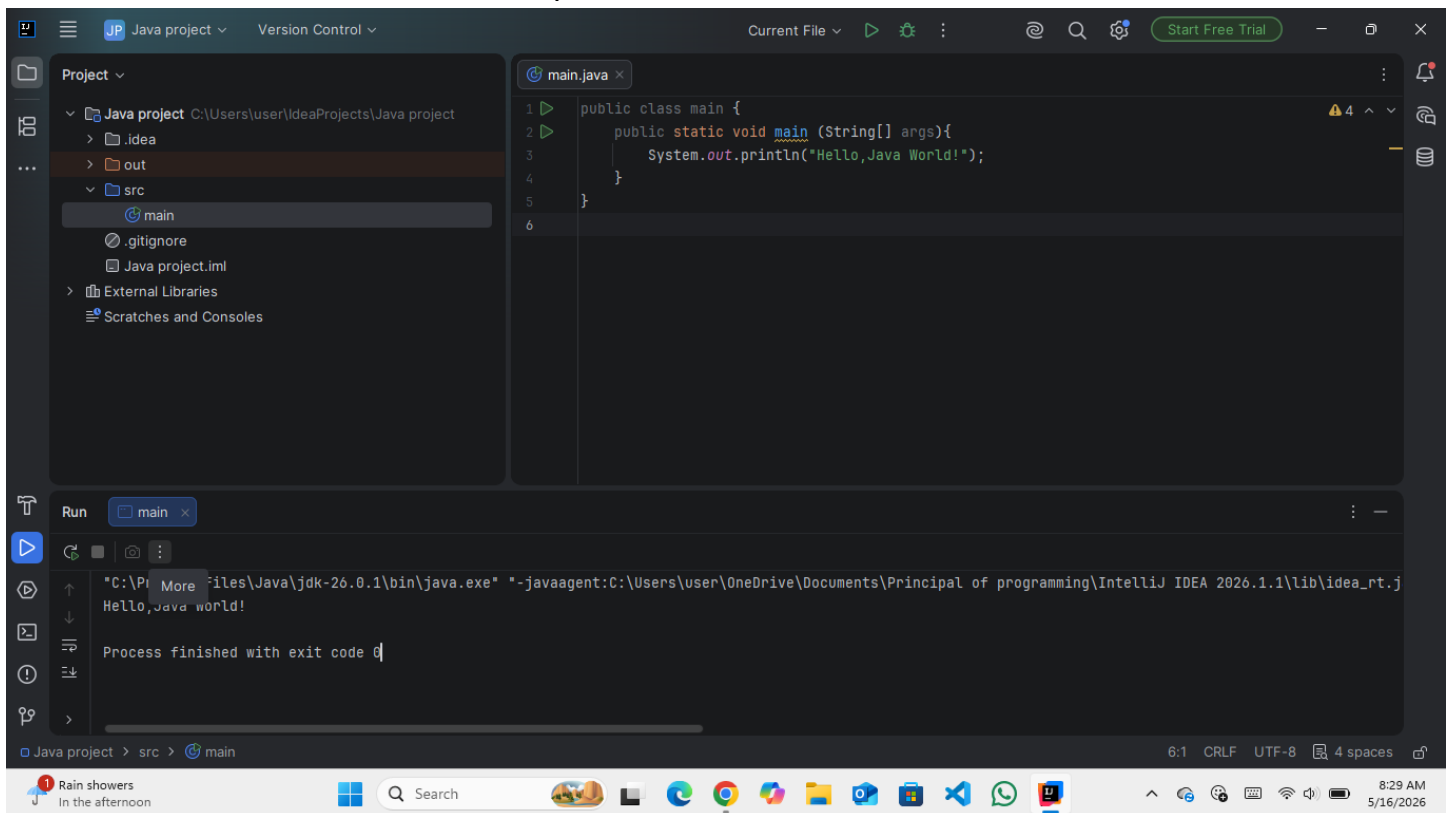
ii. Screenshot 2: Javac version check confirming Java compiler installation



2. Basic java programming task

a. Hello world program

Screenshot with output



The screenshot displays the IntelliJ IDEA IDE interface. On the left, the Project Explorer shows the file structure of a Java project, including the 'src' directory and the 'main' class file. The main editor window shows the code for 'main.java':

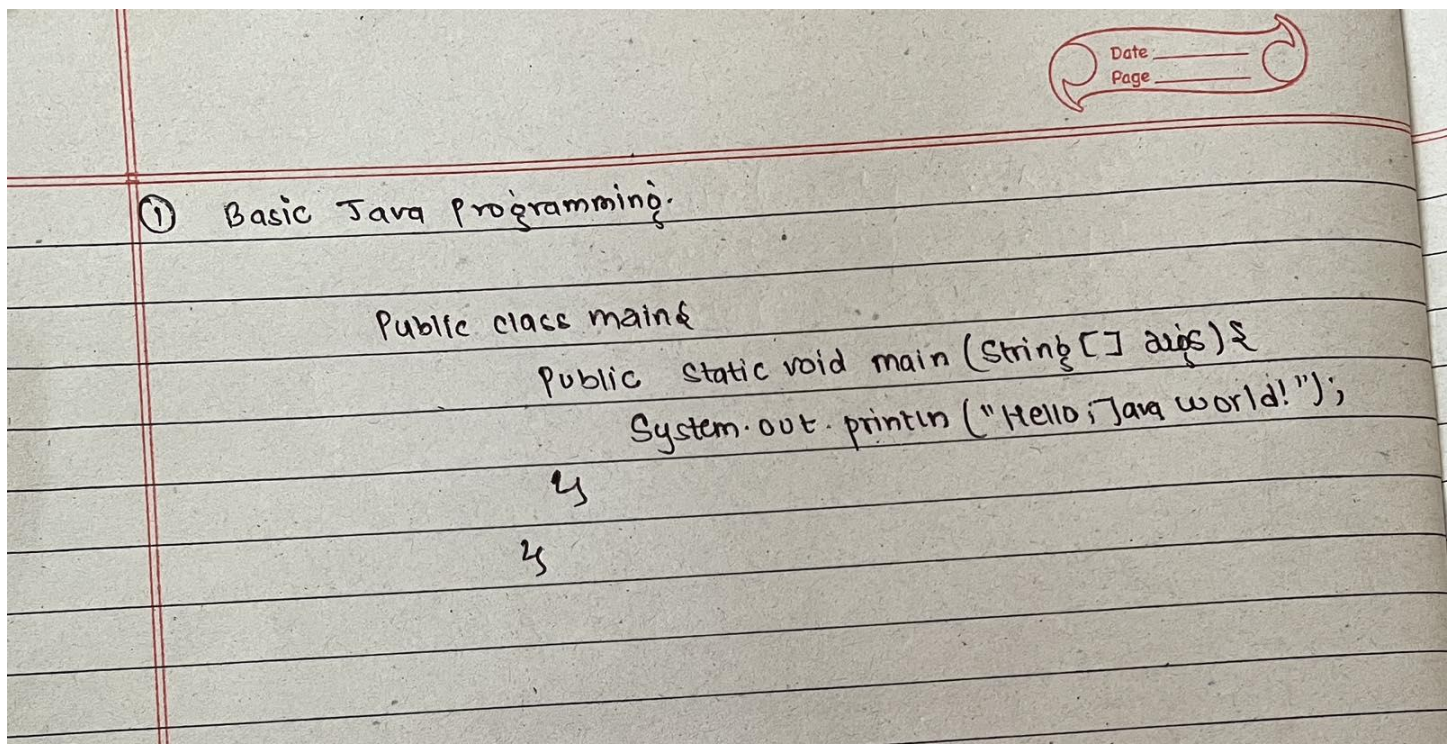
```
1 public class main {  
2     public static void main (String[] args){  
3         System.out.println("Hello,Java World!");  
4     }  
5 }  
6
```

Below the editor, the Run window shows the execution output:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\OneDrive\Documents\Principal of programming\IntelliJ IDEA 2026.1.1\lib\idea_rt.j  
Hello,Java world!  
Process finished with exit code 0
```

The status bar at the bottom indicates the file is 'main' in the 'src' directory, with a 6:1 CRLF UTF-8 4 spaces encoding.

Hand written:



The handwritten code is written on lined paper. It includes a title, a date and page field, and the Java code for a Hello World program.

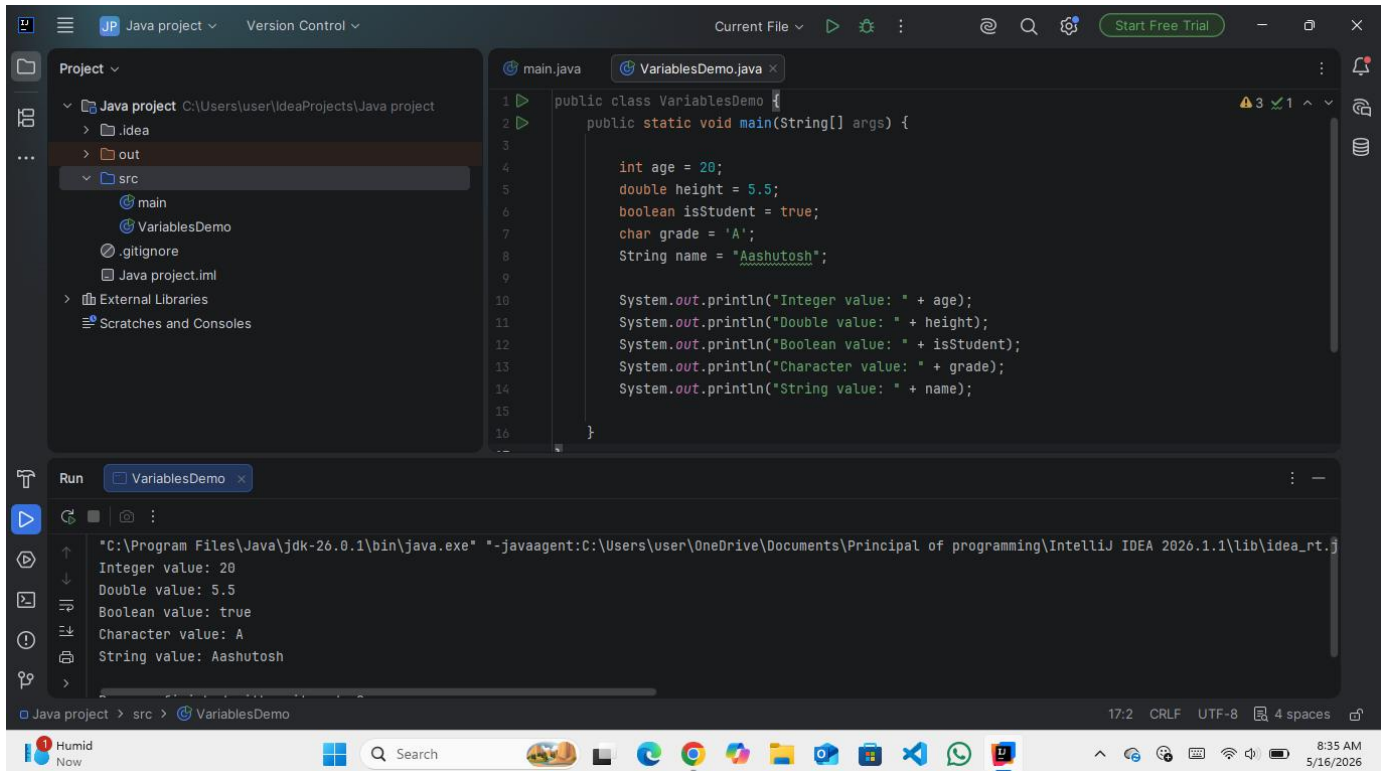
① Basic Java Programming.

Date _____
Page _____

```
Public class main {  
    Public static void main (String[] args) {  
        System.out.println ("Hello,Java world!");  
    }  
}
```


b. Variables and Data types program

Screenshots with code and output



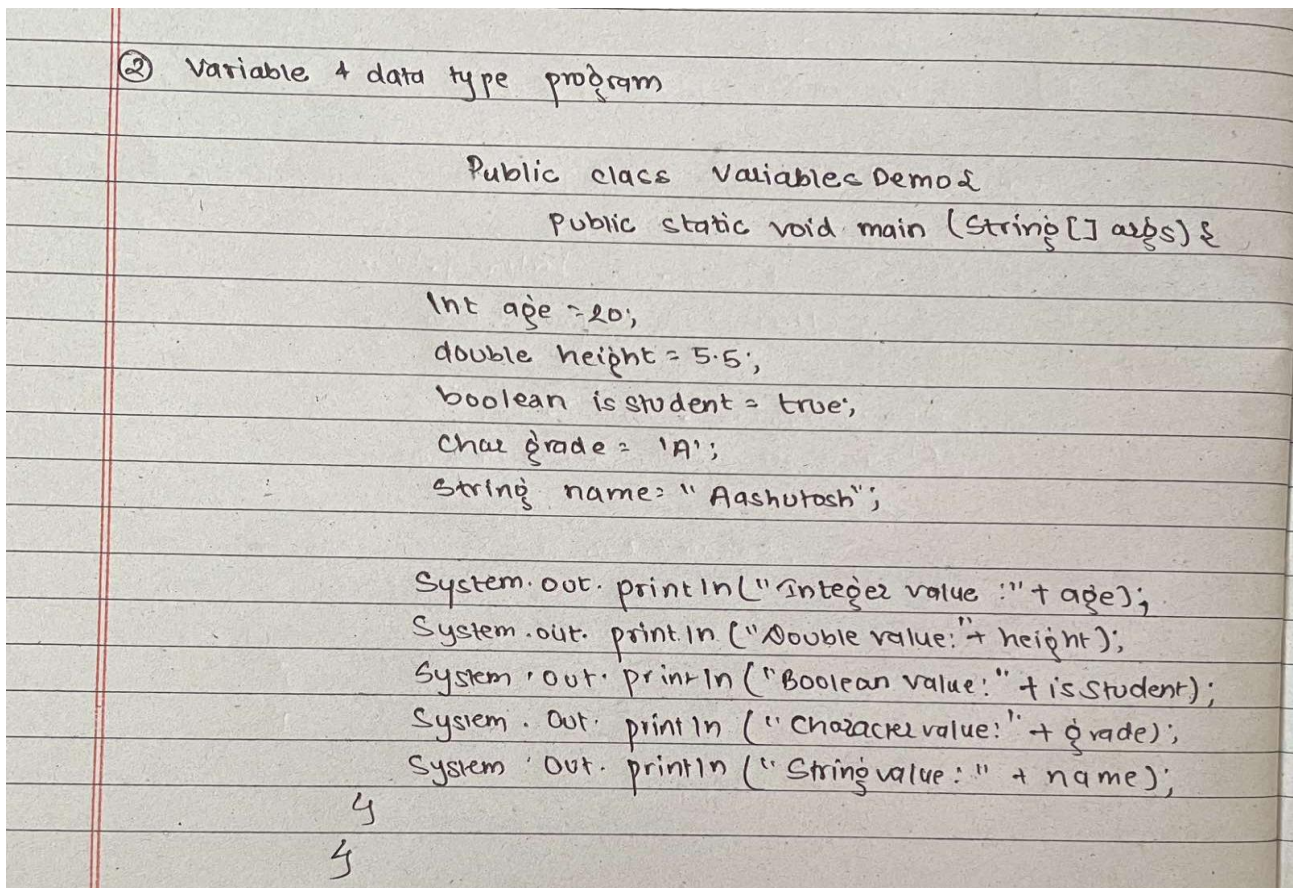
The screenshot displays the IntelliJ IDEA IDE. The left sidebar shows the project structure with 'src' containing 'main' and 'VariablesDemo'. The main editor shows the code for 'VariablesDemo.java'. The bottom panel shows the output of the program.

```
public class VariablesDemo {  
    public static void main(String[] args) {  
  
        int age = 20;  
        double height = 5.5;  
        boolean isStudent = true;  
        char grade = 'A';  
        String name = "Aashutosh";  
  
        System.out.println("Integer value: " + age);  
        System.out.println("Double value: " + height);  
        System.out.println("Boolean value: " + isStudent);  
        System.out.println("Character value: " + grade);  
        System.out.println("String value: " + name);  
    }  
}
```

Output:

```
Integer value: 20  
Double value: 5.5  
Boolean value: true  
Character value: A  
String value: Aashutosh
```

Handwritten:

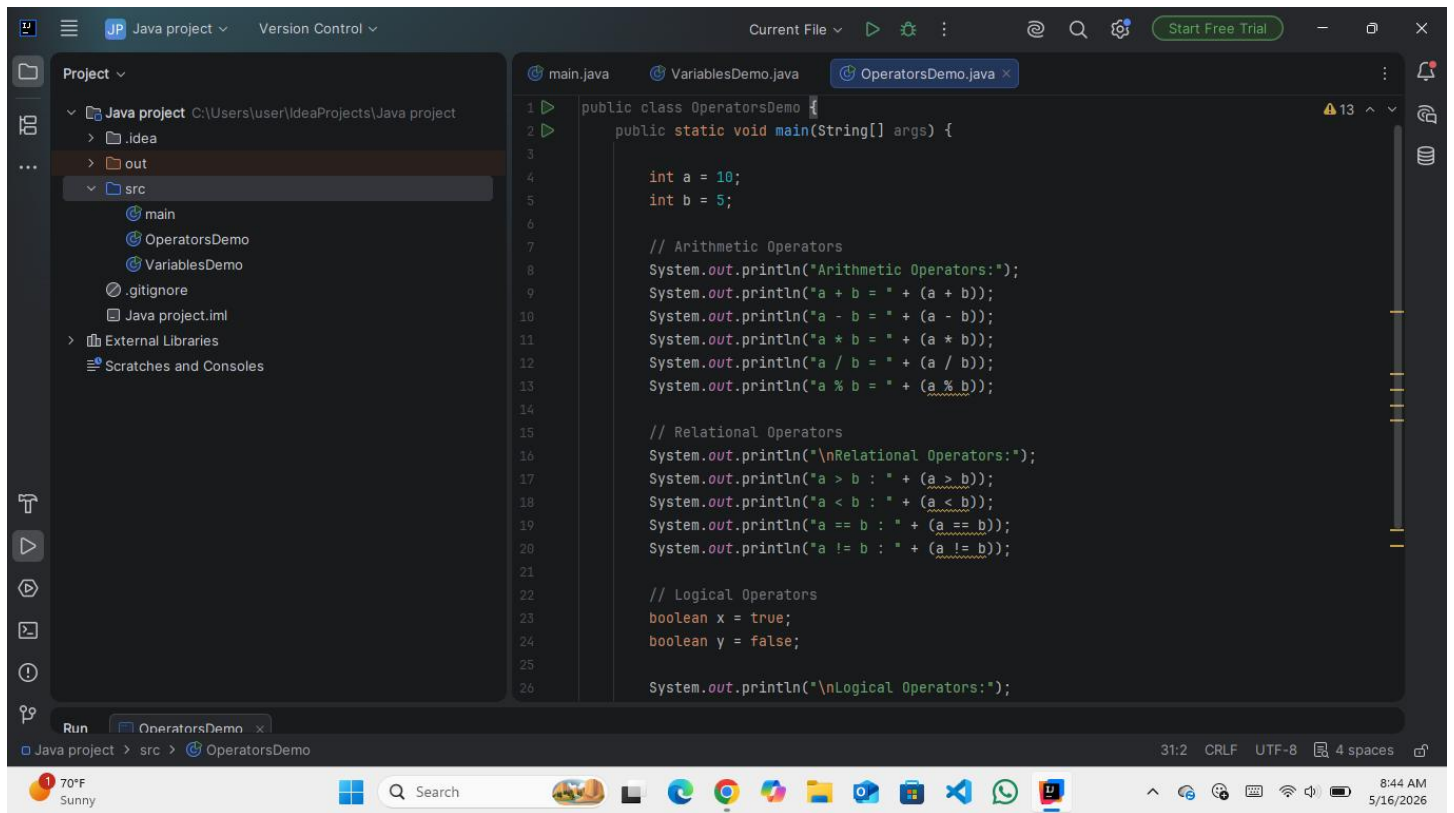


Handwritten code on lined paper:

```
② Variable + data type program  
  
Public class VariablesDemo  
    Public static void main (String[] args) {  
  
        Int age = 20;  
        double height = 5.5;  
        boolean isStudent = true;  
        Char grade = 'A';  
        String name = "Aashutosh";  
  
        System.out.println("Integer value : " + age);  
        System.out.println("Double value: " + height);  
        System.out.println("Boolean value: " + isStudent);  
        System.out.println("Character value: " + grade);  
        System.out.println("String value: " + name);  
    }  
}
```

c. Operators Program

Screenshot with code and output



The screenshot shows the IntelliJ IDEA IDE with the 'OperatorsDemo.java' file open. The code defines a public class 'OperatorsDemo' with a 'main' method. It initializes variables 'a' (10) and 'b' (5). It then demonstrates various operators: arithmetic (+, -, *, /), relational (>, <, ==, !=), and logical (true, false). The output of the program is visible in the Run console at the bottom.

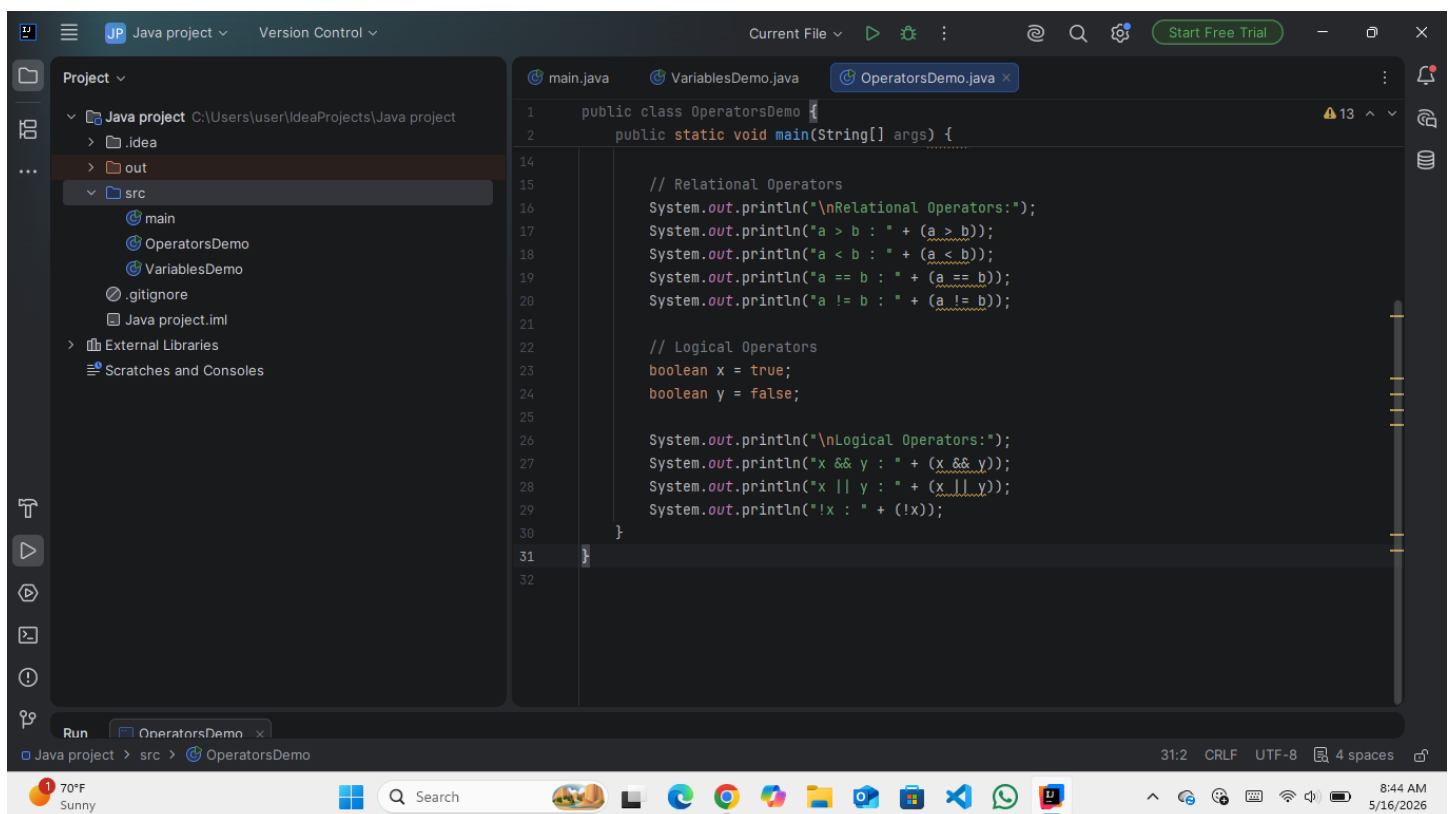
```
1 public class OperatorsDemo {
2     public static void main(String[] args) {
3
4         int a = 10;
5         int b = 5;
6
7         // Arithmetic Operators
8         System.out.println("Arithmetic Operators:");
9         System.out.println("a + b = " + (a + b));
10        System.out.println("a - b = " + (a - b));
11        System.out.println("a * b = " + (a * b));
12        System.out.println("a / b = " + (a / b));
13        System.out.println("a % b = " + (a % b));
14
15        // Relational Operators
16        System.out.println("\nRelational Operators:");
17        System.out.println("a > b : " + (a > b));
18        System.out.println("a < b : " + (a < b));
19        System.out.println("a == b : " + (a == b));
20        System.out.println("a != b : " + (a != b));
21
22        // Logical Operators
23        boolean x = true;
24        boolean y = false;
25
26        System.out.println("\nLogical Operators:");
```

Run console output:

```
Arithmetic Operators:
a + b = 15
a - b = 5
a * b = 50
a / b = 2
a % b = 0

Relational Operators:
a > b : true
a < b : false
a == b : false
a != b : true

Logical Operators:
x && y : false
x || y : true
!x : false
```



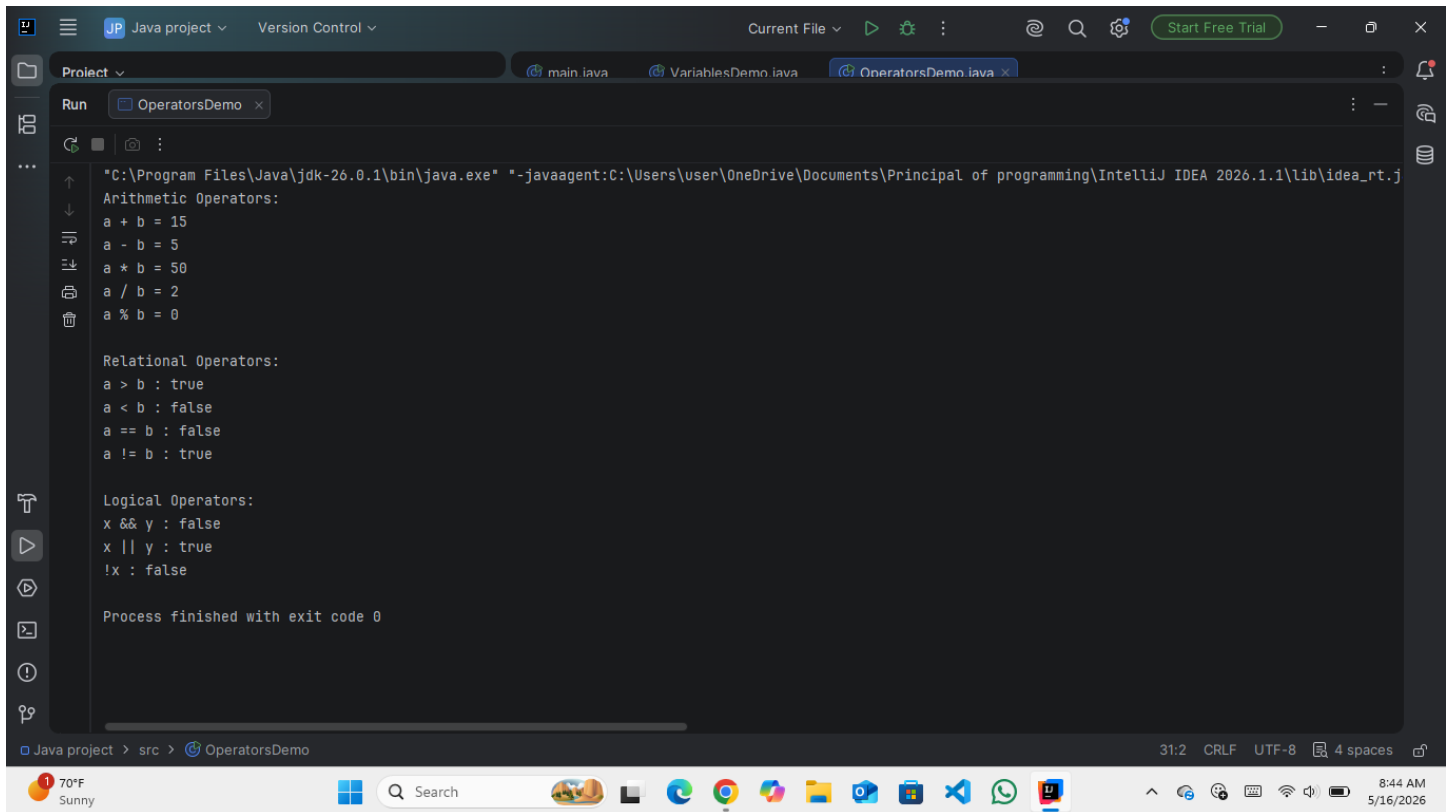
The screenshot shows the IntelliJ IDEA IDE with the 'OperatorsDemo.java' file open. The code defines a public class 'OperatorsDemo' with a 'main' method. It demonstrates various operators: relational (>, <, ==, !=) and logical (true, false). The output of the program is visible in the Run console at the bottom.

```
1 public class OperatorsDemo {
2     public static void main(String[] args) {
3
4         // Relational Operators
5         System.out.println("\nRelational Operators:");
6         System.out.println("a > b : " + (a > b));
7         System.out.println("a < b : " + (a < b));
8         System.out.println("a == b : " + (a == b));
9         System.out.println("a != b : " + (a != b));
10
11        // Logical Operators
12        boolean x = true;
13        boolean y = false;
14
15        System.out.println("\nLogical Operators:");
16        System.out.println("x && y : " + (x && y));
17        System.out.println("x || y : " + (x || y));
18        System.out.println("!x : " + (!x));
19    }
20 }
```

Run console output:

```
Relational Operators:
a > b : true
a < b : false
a == b : false
a != b : true

Logical Operators:
x && y : false
x || y : true
!x : false
```

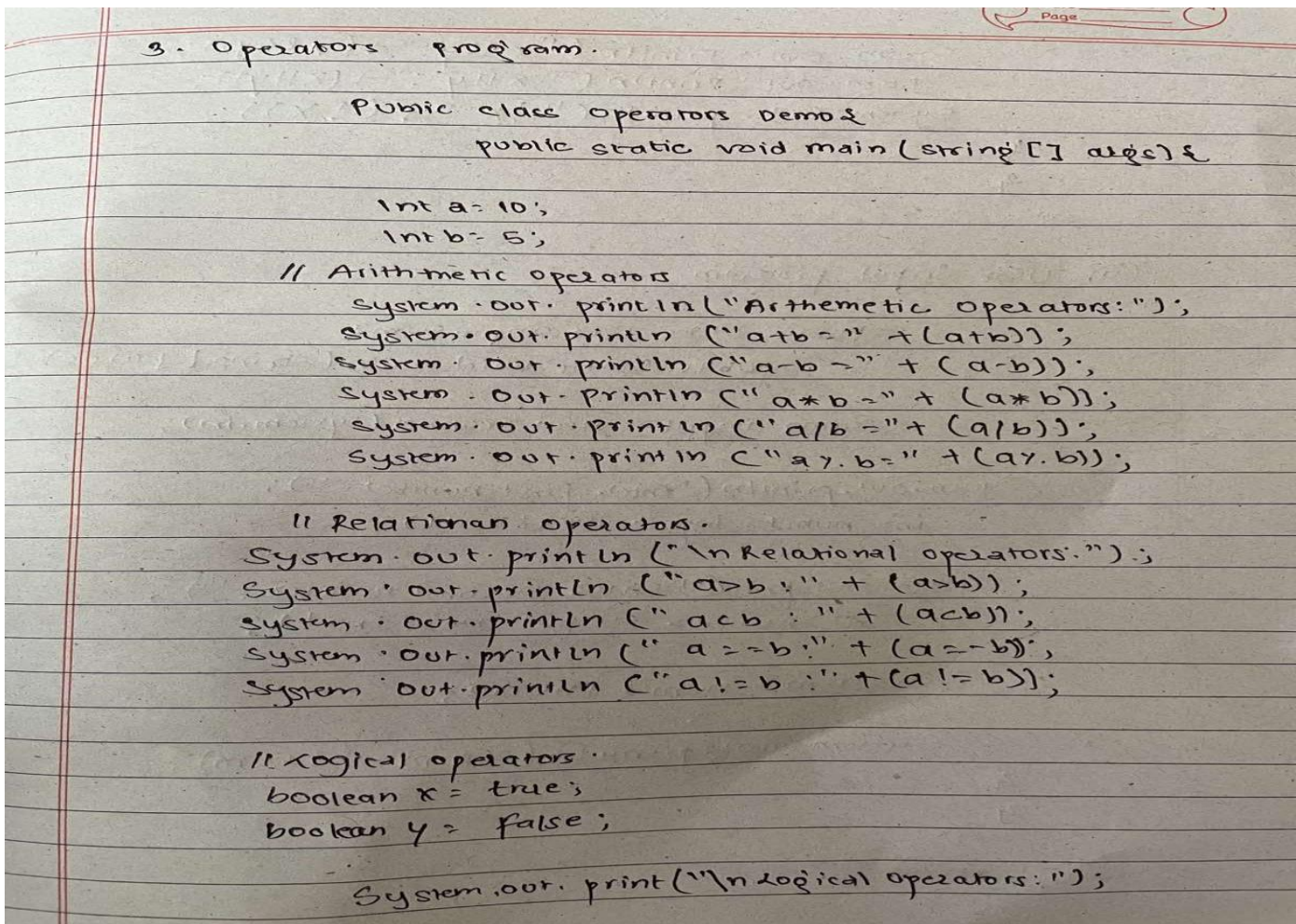
```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\OneDrive\Documents\Principal of programming\IntelliJ IDEA 2026.1.1\lib\idea_rt.j
Arithmetic Operators:
a + b = 15
a - b = 5
a * b = 50
a / b = 2
a % b = 0

Relational Operators:
a > b : true
a < b : false
a == b : false
a != b : true

Logical Operators:
x && y : false
x || y : true
!x : false

Process finished with exit code 0
```

Handwritten:



```
3. Operators program.

Public class operators demo {
    public static void main (String [] args) {

        int a = 10;
        int b = 5;

        // Arithmetic operators
        System.out.println ("Arithmetic Operators:");
        System.out.println ("a+b = " + (a+b));
        System.out.println ("a-b = " + (a-b));
        System.out.println ("a*b = " + (a*b));
        System.out.println ("a/b = " + (a/b));
        System.out.println ("a % b = " + (a % b));

        // Relational operators.
        System.out.println ("\n Relational operators.");
        System.out.println ("a > b : " + (a > b));
        System.out.println ("a < b : " + (a < b));
        System.out.println ("a == b : " + (a == b));
        System.out.println ("a != b : " + (a != b));

        // Logical operators.
        boolean x = true;
        boolean y = false;

        System.out.println ("\n Logical operators:");
```

Date _____
Page _____

```
System.out.println("x & y:" + (x & y));  
System.out.println("x || y:" + (x || y));  
System.out.println("!x:" + (!x));
```

```
{  
}  
}
```

d. User input program

Screenshot of code

The screenshot displays the IntelliJ IDEA IDE with a Java project. The project structure on the left shows a source folder with files: main, OperatorsDemo, UserInputSum, and VariablesDemo. The main editor window shows the code for UserInputSum.java, which is a public class with a main method. The code prompts the user for two numbers, calculates their sum, and prints the result. The Run window at the bottom shows the execution output, where the user entered 10 and 30, resulting in a sum of 40. The status bar at the bottom indicates the build is successful and the file encoding is UTF-8.

```
public class UserInputSum {  
    public static void main(String[] args) {  
  
        Scanner input = new Scanner(System.in);  
  
        System.out.print("Enter first number: ");  
        int num1 = input.nextInt();  
  
        System.out.print("Enter second number: ");  
        int num2 = input.nextInt();  
  
        int sum = num1 + num2;  
  
        System.out.println("Sum = " + sum);  
    }  
}
```

Run UserInputSum

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Users\user\OneDrive\Documents\Principal of programming\IntelliJ IDEA 2026.1.1\lib\idea_rt.j  
Enter first number: 10  
Enter second number: 30  
Sum = 40  
Process finished with exit code 0
```


(u) User Input program.

```
public class User Input Sum {  
    public static void main (String [] args) {  
        Scanner input = new Scanner (System.in);  
        System.out.println ("Enter first number:");  
        int num1 = input.nextInt();  
        System.out.println ("Enter second number:");  
        int num2 = input.nextInt();  
        int sum = num1 + num2;  
        System.out.println ("Sum = " + sum);  
    }  
}
```

All code in git : <https://github.com/AashutoshSubedi/java-programs>

3. Git & Git hub certificate

